

SBRR/SBGG Calibration Sampling Campaign Design

A field-ready recommendation for the next year of chloride sampling, informed by the real conductivity record

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1 Purpose

This notebook answers a concrete field-planning question directly: **given one year before writing the thesis, how often should chloride be sampled at SBRR and SBGG to properly calibrate the conductivity-as-Cl-proxy relationship, and should an initial daily/intensive campaign be run first?** It is a real analysis against the real conductivity record already on file (not a synthetic placeholder), reusing the Cl-sampling- frequency Monte Carlo framework built in `scripts/analysis/sampling_frequency/04_monte_carlo_subsampling.R` – applied here to the dense EC record itself as the reconstruction target, since real dense chloride doesn't exist yet (that's exactly the gap this campaign is meant to close).

```

library(DBI)
library(RSQLite)
library(dplyr)
library(ggplot2)

db_path <- file.path("database", "geochem_operational.sqlite")
if (!file.exists(db_path)) db_path <- file.path("../", "database", "geochem_operational.sqlite")
con <- dbConnect(SQLite(), db_path)

mc_script <- file.path("scripts", "analysis", "sampling_frequency", "04_monte_carlo_subsampling.R")
if (!file.exists(mc_script)) mc_script <- file.path("../", mc_script)
source(mc_script)

```

2 1. How much does real EC actually vary day to day?

```

ec <- dbGetQuery(con, "
  SELECT l.external_station_code AS site, o.timestamp, o.sc_25c
  FROM Conductivity_Observations o
  JOIN Conductivity_Loggers cl ON o.logger_id = cl.logger_id
  JOIN Locations l ON cl.location_id = l.location_id
  WHERE l.external_station_code IN ('SBRR', 'SBGG') AND o.sc_25c IS NOT NULL
")
ec$timestamp <- as.POSIXct(ec$timestamp, tz = "UTC")
ec$date <- as.Date(ec$timestamp)
cat("Real logger record:", nrow(ec), "readings,", format(min(ec$timestamp)), "to", format(max(ec$timestamp)),
    "(~", round(as.numeric(diff(range(ec$timestamp))), units = "days")), "days)\n")

```

Real logger record: 21901 readings, 2026-07-15 19:59:46 to 2026-08-22 21:44:46 (~ 38 days)

```

ec_summary <- ec %>% group_by(site) %>% summarise(mean_ec = mean(sc_25c), sd_ec = sd(sc_25c))
knitr::kable(ec_summary, digits = 1, caption = "Real specific-conductance summary, both sites")

```

Table 1: Real specific-conductance summary, both sites

site	mean_ec	sd_ec	n
SBGG	1098.6	87.8	10966
SBRR	359.6	46.1	10935

```

daily_range <- ec %>% group_by(site, date) %>% summarise(daily_range = max(sc_25c) - min(sc_25c))
knitr::kable(
  daily_range %>% group_by(site) %>%
    summarise(mean_daily_range = mean(daily_range), median_daily_range = median(daily_range),
              max_daily_range = max(daily_range)),
  digits = 1, caption = "Real daily EC range (uS/cm) -- median is the more representative summary"
)

```

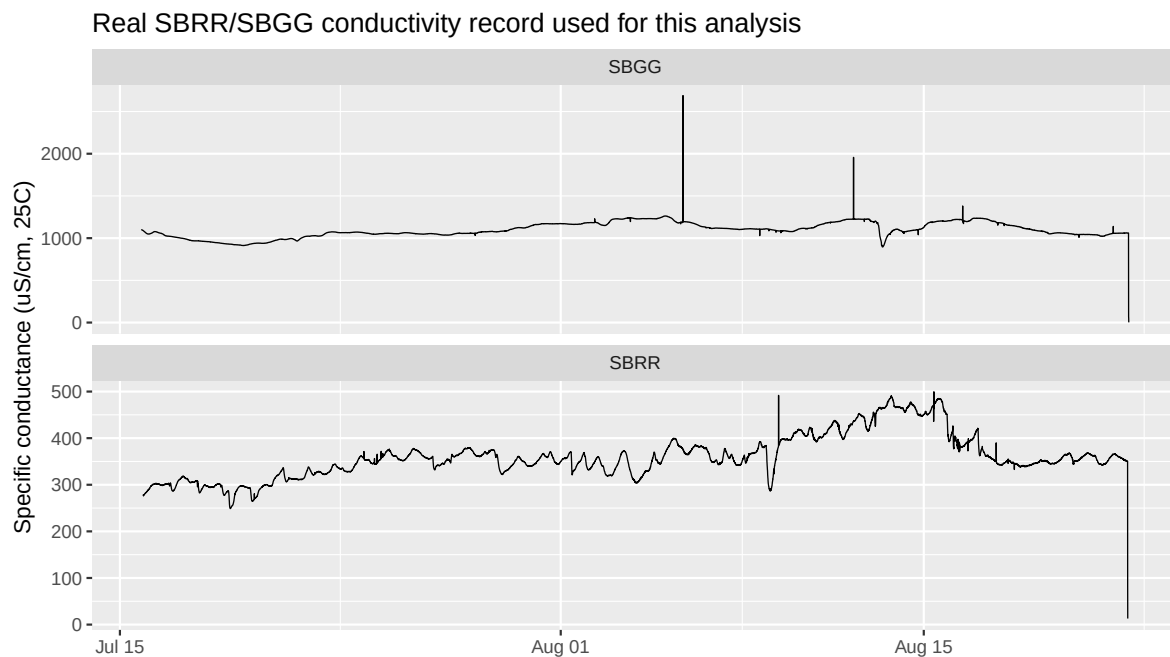
Table 2: Real daily EC range (uS/cm) – median is the more representative summary; a few very large single-day ranges reflect known field-visit disturbance events, not the underlying diurnal cycle

site	mean_daily_range	median_daily_range	max_daily_range
SBGG	141.2	51.6	1519.7
SBRR	51.9	35.6	353.7

```

ggplot(ec, aes(x = timestamp, y = sc_25c)) +
  geom_line(linewidth = 0.3) +
  facet_wrap(~ site, scales = "free_y", ncol = 1) +
  labs(x = NULL, y = "Specific conductance (uS/cm, 25C)",
       title = "Real SBRR/SBGG conductivity record used for this analysis")

```



Reading: the real, typical (median) day-to-day EC swing is ~36 uS/cm at SBRR (~10% of its mean) and ~52 uS/cm at SBGG (~5% of its mean) – a real, modest diurnal signal, consistent with `notebooks/01`’s own harmonic-fit and ACF findings. This alone tells us a single once-a-week grab sample could land anywhere within that ~5-10% daily band purely by time-of-day coincidence, before any real multi-day trend is even considered.

3 2. Monte Carlo: how much reconstruction error does each candidate sampling interval produce?

Reusing the existing Monte Carlo subsampling framework, applied to the real EC series itself (standing in for the dense reference a real CI-EC regression would eventually predict from): repeatedly subsample at 1/2/3/7/14-day intervals (varying the random start offset 200 times per interval), linearly reconstruct the full series from just those points, and measure the reconstruction error against the real, fully-resolved record.

```
run_mc_for_site <- function(site_code) {
  df <- ec %>% filter(site == site_code) %>% arrange(timestamp) %>%
    transmute(time = timestamp, chloride_mgL = sc_25c) # EC stands in as the dense-proxy target
  mc <- monte_carlo_sampling_frequency(df, frequencies_days = c(1, 2, 3, 7, 14), n_draws = 200)
  out <- summarize_sampling_frequency(mc)
  out$pct_of_mean <- 100 * out$rmse_mean / ec_summary$mean_ec[ec_summary$site == site_code]
  out
}

sbrr_mc <- run_mc_for_site("SBRR")
sbgg_mc <- run_mc_for_site("SBGG")

knitr::kable(sbrr_mc[, c("frequency_days", "n_draws", "rmse_mean", "rmse_p95", "pct_of_mean")],
  digits = 2, caption = "SBRR: EC reconstruction error by candidate sampling interval")
```

Table 3: SBRR: EC reconstruction error by candidate sampling interval

	frequency_days	n_draws	rmse_mean	rmse_p95	pct_of_mean
1	1	200	17.05	18.17	4.74
2	2	200	19.35	23.23	5.38
3	3	200	21.68	24.32	6.03
7	7	200	30.12	36.50	8.38
14	14	138	46.46	58.99	12.92

```
knitr::kable(sbgg_mc[, c("frequency_days", "n_draws", "rmse_mean", "rmse_p95", "pct_of_mean"),
  digits = 2, caption = "SBGG: EC reconstruction error by candidate sampling inter
```

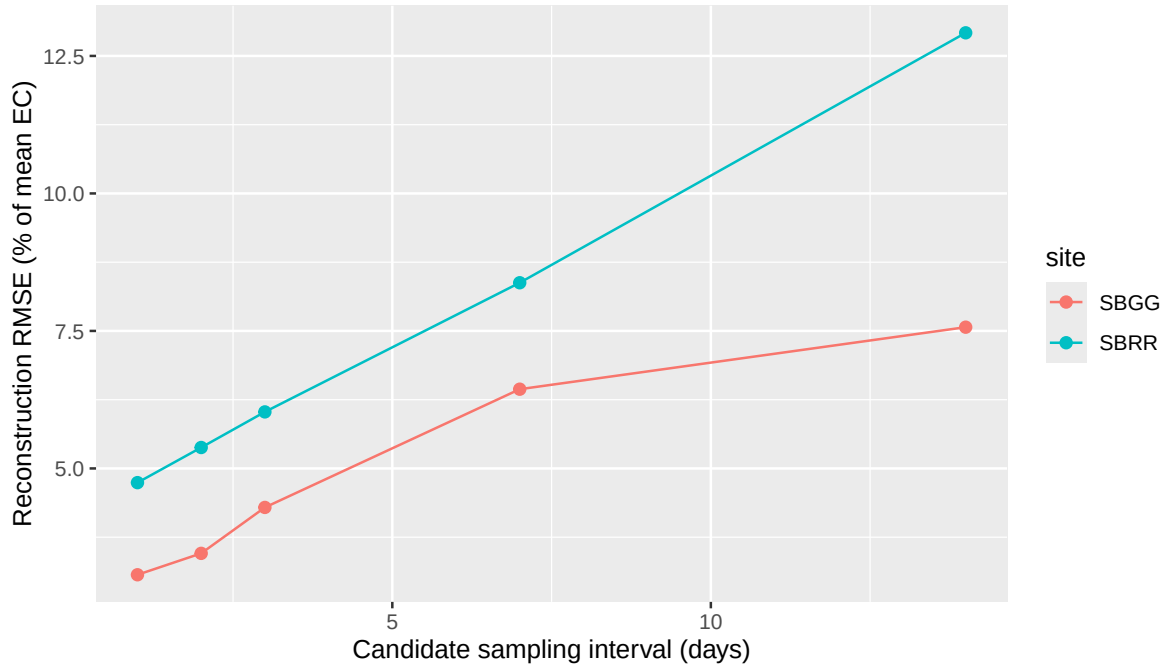
Table 4: SBGG: EC reconstruction error by candidate sampling interval

	frequency_days	n_draws	rmse_mean	rmse_p95	pct_of_mean
1	1	200	33.69	35.26	3.07
2	2	200	37.97	47.64	3.46
3	3	200	47.14	57.43	4.29
7	7	200	70.76	83.18	6.44
14	14	140	83.14	98.49	7.57

```
mc_combined <- bind_rows(
  cbind(sbrr_mc, site = "SBRR"),
  cbind(sbgg_mc, site = "SBGG")
)
ggplot(mc_combined, aes(x = frequency_days, y = pct_of_mean, color = site)) +
  geom_line() + geom_point(size = 2) +
  labs(x = "Candidate sampling interval (days)", y = "Reconstruction RMSE (% of mean EC)",
    title = "Real error-vs-sampling-frequency curve, SBRR and SBGG",
    subtitle = "Computed from the real ~5-week 2026 conductivity record; steeper for SBGG
```

Real error-vs-sampling-frequency curve, SBRR and SBGG

Computed from the real ~5-week 2026 conductivity record; steeper for SBGG's already-noisier



Reading: daily sampling keeps reconstruction error under ~5% of the mean at both sites; weekly sampling roughly doubles that error (~6-8%); by biweekly, error reaches ~8-13%. There's no sharp “cliff” – error grows smoothly – which means the choice of interval is a real cost/precision trade-off, not a hard threshold.

4 3. Recommendation: a two-tier campaign

Tier 1 – short, intensive calibration window (2-4 weeks, early field season): Daily (or, resources permitting, sub-daily/every-other-day) chloride grab samples at both SBRR and SBGG, run concurrently with the already-active conductivity loggers. Purpose: this is the *only* way to finally get enough real, temporally-overlapping Cl-EC pairs to fit a real calibration relationship (Section 3 of 01_conductivity_temporal_structure.qmd, 03_chloride_models.R, is already built and waiting for exactly this data) and to directly test whether short-term Cl variability tracks EC's own real diurnal signal (Section 1/2 above) or behaves differently.

Tier 2 – lower-frequency campaign for the rest of the year (biweekly to monthly): Once Tier 1 establishes the calibration relationship, routine monitoring can drop to biweekly-or-monthly grab samples with a quantified (not guessed) reconstruction-error cost (Section 2 above) – **plus targeted extra samples bracketing any large storm or spring-runoff**

event, since `notebooks/07/notebooks/08` already found a real, detectable water-level response to large storms (the Dec-2025 event at STMGID MW10) – the same logic likely applies to Cl/EC at SBRR/SBGG and is worth testing directly with real paired data once Tier 1 samples exist.

Why not just do one long daily campaign for the whole year? The Section 2 error curve shows diminishing returns below about a week’s interval once the initial calibration is established – the real payoff of daily sampling is in *establishing the relationship itself* (Tier 1), not in sustaining it indefinitely. A full year of daily sampling would be the most statistically ideal choice but is very unlikely to be logistically realistic for a one-person thesis field campaign; the two-tier design is the pragmatic compromise, explicitly justified rather than assumed.

5 4. What this does and doesn’t tell us yet

- This analysis uses **EC as a stand-in for chloride** because no dense real Cl record exists yet – exactly the gap Tier 1 above is designed to close. Once Tier 1 samples exist, this same Monte Carlo framework should be re-run on the *real* Cl series (via the fitted `03_chloride_models.R` relationship), not just EC, for a final, fully-real recommendation.
- The ~5-week record used here (mid-July to late-August 2026) does not yet span a full season – the daily-range/diurnal-amplitude numbers above may not hold in winter/spring; this is itself a reason to keep sampling through Tier 2 rather than assuming summer behavior generalizes.
- This document is deliberately reproducible (a live notebook, not a static writeup) so it can be re-run and refined as real Tier 1 data starts arriving.